

5 Dopamine, serotonin and acetylcholine are neurotransmitters in the human nervous system.

(a) Describe how a neurotransmitter allows an impulse to be transferred to an adjacent neurone.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) The diagram shows how serotonin is produced from tryptophan.



Enzymes A and B are found in the Madagascar periwinkle.

Medical serotonin can be produced by genetically engineered *E.coli* bacteria.

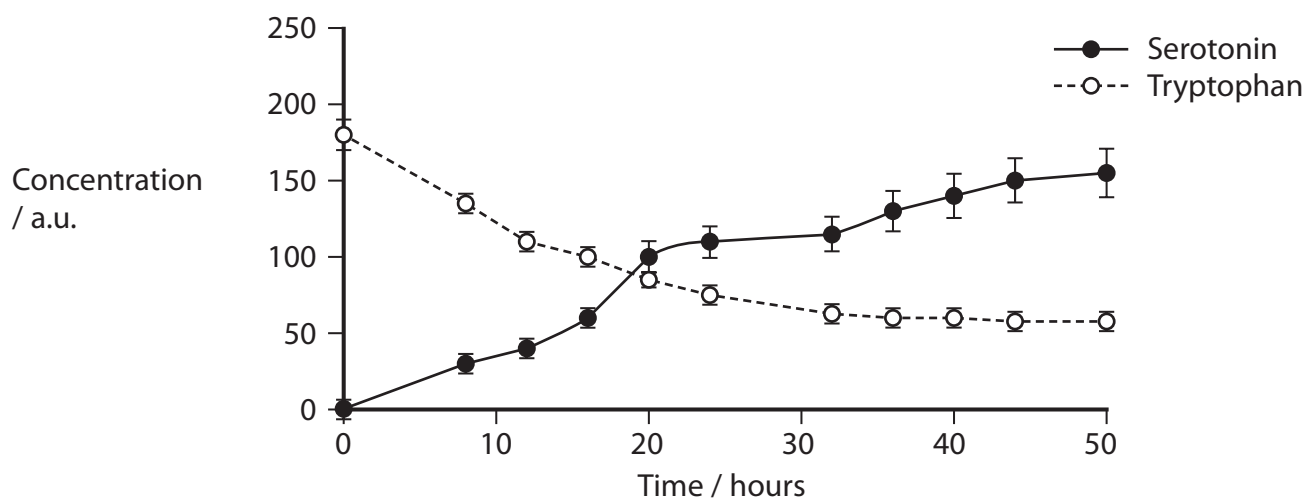
Describe how *E.coli* could be genetically engineered using the Madagascar periwinkle to produce serotonin.

(4)

[illegible]

- (c) The genetically engineered *E. coli* was grown in a liquid culture.
The concentrations of tryptophan and serotonin in the culture were measured.

The graph shows the results.



Discuss the changes shown in the graph.

Use the information in the question to support your answer.

(4)

(Total for Question 5 = 12 marks)